Question	Answer	Marks	Partial Marks
4(b)(i)	$420 < d \leqslant 450$	**1**	
4(b)(ii)	411.25	**4**	**M1** for 275, 350, 410, 435, 475 soi **M1** for Σfx **M1 dep** for *their* $\Sigma fx \div 80$
4(b)(iii)	2.6 19 14	**3**	**B1** for each If 0 scored, **SC1** for 3 of 0.14, 0.13, 0.95 or 0.7 oe
4(b)(iv)	$\frac{7}{158}$ oe	**3**	**M2** for $[2\times]\frac{20}{80}\times\frac{7}{79}$ oe or **M1** for $\frac{20}{80}$ or $\frac{7}{79}$ or $\frac{7}{80}$ or $\frac{20}{79}$ oe seen After 0 scored, **SC1** for $\frac{7}{160}$ oe
5(a)(i)	$\begin{pmatrix} 4 \\ -12 \end{pmatrix}$	**2**	**B1** for each
5(a)(ii)	$1^2 + 7^2$	**M1**	
	$5^2 + ([-]5)^2$	**M1**	
	Both $\sqrt{50}$ oe	**A1**	With no errors seen If M0M0A0 scored SC1 for $\sqrt{50}$ oe for each
5(a)(iii)	44.4 or 44.42[8…] to 44.435	**2**	FT *their* **(a)(ii)** correct to 3sf or better **M1** for $2 \times \pi \times$ *their* $\sqrt{50}$ oe
5(a)(iv)	(3, 1)	**2**	**B1** for each

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Question	Answer	Marks	Partial Marks
5(a)(v)	[y =] $\frac{1}{3}x$	**4**	**B3** for a correct equation in the wrong form as final answer Or **B2** for 1/3 stated or used as perpendicular gradient OR **M1** for [grad PQ] = $\frac{7--5}{1-5}$ oe **M1** for $\frac{-1}{\textit{their}\ \text{grad}\ PQ}$ **M1dep** for substituting *their***(a)(iv)** or (0,0) into y = *their* $mx + c$ oe dep on the 2nd M1 or B2
5(b)	$\frac{3}{5}$ **a** + $\frac{2}{5}$ **b** final answer	**4**	**B3** for an unsimplified correct answer or **B2** for $AM = \frac{2}{5}(\boldsymbol{b}-\boldsymbol{a})$ soi or $BM = \frac{3}{5}(\boldsymbol{a}-\boldsymbol{b})$ soi or **B1** for AB = **b – a** or BA = **a – b** or for a correct route for OM or for correct diagram
6(a)	245	**1**	
6(b)(i)	180 – (55 + 25) [=100]	**M1**	
6(b)(ii)	$\frac{32\times\sin 25}{\sin 100}$ oe	**M2**	**M1** for $\frac{\sin 25}{BH} = \frac{\sin 100}{32}$ oe
	13.73…	**A1**	